

MEKDELAAMBAUNIVERSITY

College Of Computing and Informatics Department Of Software Engineering

Project Title:Smart Personal Dashboard

Course Name : Software Project Management

Course code :SEngb032

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SubmissionDate:15/08/2018 E.C

Acknowledgement

First and foremost, we would like to thank the Almighty God for giving us the strength, health, and patience to complete this project successfully.

We would like to express our deepest gratitude to our instructor for his valuable guidance, constructive feedback, and continuous support throughout this project. His expertise in project management and software engineering helped us to shape this work properly.

We are also grateful to our families and friends for their encouragement and understanding during the long hours we spent working on this project. Their moral support kept us motivated.

Finally, we would like to thank all stakeholders and users who provided their feedback during the requirement gathering and testing phases. Their input was essential in making this project practical and user-centered.

Abstract

The Smart Personal Finance Dashboard is a web-based software solution designed to help individuals, especially students and young professionals, track, analyze, and optimize their personal spending and saving habits. The project addresses the problem of "financial blindness," where people are unaware of their small daily expenses until they face a budget deficit. The system includes features such as income and expense tracking, category-based budgeting, visual analytics (pie charts and bar graphs), budget alerts, and historical reporting. The project follows standard project management methodologies, including a Project Charter, Work Breakdown Structure (WBS), Gantt Chart, Critical Path Analysis (CPA), Cost Estimation, and Risk Register. The total estimated budget is 1,456,455 ETB, and the project duration is 6 weeks (24 days on the critical path). This dashboard empowers users to take control of their financial health through an intuitive and data-driven platform.

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Chapter one(1):

Introduction

1.1 Project Overview

The Smart Personal Finance Dashboard is a comprehensive software solution designed to address the growing challenges of individual financial management in the modern era. As digital transactions become more frequent, tracking where every cent goes has become an overwhelming task for many. This project aims to simplify this process by developing an intuitive, data-driven platform that allows users to monitor, analyze, and optimize their personal spending and saving habits.

1.2 Background and Motivation

In the current economic climate, financial literacy and disciplined money management are essential skills for personal stability. Many people, especially students and young professionals, suffer from "financial blindness"—a state where they are unaware of their cumulative small expenditures until they face a budget deficit at the end of the month. The motivation behind this project is to leverage software engineering methodologies to create a tool that moves beyond simple record-keeping, providing users with actionable insights into their financial behavior.

1.3 Problem Statement

The primary problem addressed by this project is the lack of a centralized, automated, and easy-to-use financial tracking system. Traditional methods suffer from several drawbacks:

- Manual Effort: Notebooks and simple spreadsheets require constant manual entry, which is prone to forgetfulness and errors.
- Lack of Visualization: Raw numbers are difficult to interpret; without charts and graphs, users cannot easily identify spending trends. • Fragmented Data: Financial information is often spread across various bank messages, receipts, and cash transactions, making it hard to see the "big picture."

1.4 Objectives of the System

The specific objectives of the Smart Personal Finance Dashboard are:

1. Centralized Tracking: To provide a single platform where all income and expenses can be recorded and categorized (e.g., Housing, Transportation, Utilities, Entertainment).

2. Visual Analytics: To implement graphical representations such as pie charts and bar graphs that provide an immediate visual summary of financial health.

3. Budgetary Control: To enable users to set monthly limits for specific categories and receive alerts when they approach these limits.

4. Historical Reporting: To maintain a secure database of past transactions, allowing users to compare their financial growth over months or years.

1.5 Scope and Constraints

• Scope: The project covers the design and development of the user interface, backend database architecture, and logic for financial calculations. It includes user authentication (Login/Signup) and a personalized dashboard.

• Constraints: The initial version will be a standalone desktop or web application. It will rely on user-inputted data rather than direct bank API integration to ensure privacy and simplify the initial development phase within the academic semester timeframe.

Chapter 2:

Project Charter and Scope Statement & WBS

2.1 Project Charter

2.1.1 Problem Statement

Currently, users rely on multiple disconnected apps and websites to track their schedules, health, finances, and social interactions. This fragmentation leads to time wasted switching contexts, missed notifications, and poor data-driven decision-making. The Smart Personal Dashboard project aims to solve this problem by providing a single-pane-of-glass view of the user's digital life.

2.1.2 SMART Objectives

- Specific: Develop a web-based dashboard that aggregates income, expenses, and savings data.
- Measurable: Track at least 5 expense categories, support 10+ customizable budget alerts, and achieve 100 active users in the first month post-launch.
- Achievable: Using a team of 3 developers and agile methodologies, the MVP can be built in 6 weeks.
- Relevant: Addresses the growing market for personal productivity and personal finance management tools.
- Time-bound: Complete MVP deployment within 6 weeks.

2.1.3 Key Stakeholders

	A	B	C
1	**Key Stakeholders**		
2	**Stakeholder**	**Role**	**Interest**
3	Project Sponsor	Provides funding and strategic direction	High
4	Project Manager	Oversees planning, execution, and delivery	High
5	Development Team	Designs, codes, and tests the software	High
6	End Users	Use the dashboard for personal management	Medium
7	Instructor (Dr. Banteamlak)	Evaluates the assignment deliverables	Medium
8	API Providers (Google, Fitbit, etc.)	Provide data through APIs	Low

Table 1.1

2.1.4 High-Level Budget

The total estimated budget for the Smart Personal Finance Dashboard project is 1,456,455 ETB.. This financial plan is structured to cover all necessary resources for a successful implementation:

- Labor: The primary expenditure is 1,040,000, which covers 160 person-hours of professional development and management at an hourly rate of 6,750 ETB.
- Software and Licenses: A total of 89,050 ETB is allocated for essential cloud hosting services (such as AWS) and professional software licenses required for design and project management.
- Hardware: 195,000 ETB is set aside for the procurement of testing devices, including laptops and smartphones, to ensure system compatibility.
- Contingency Reserve: A dedicated fund of 132,405 ETB (representing 10% of the total budget) is maintained to address any unforeseen technical challenges or risks that may arise during the project lifecycle.

2.1.5 Key Milestones

Milestone ID	Deliverable/Milestone	Target Date	Description
M1	**Project Charter Approved**	**Week 1**	**Finalization of SMART objectives, stakeholder identification**
M2	**Requirements & Design Complete**	**Week 2**	**Completion of Scope Statement, WBS, and UI/UX wireframes**

M3	**Core API Integrations**	**Week 3**	**Implementation of backend logic for transaction handling**
M4	**MVP Development Complete**	**Week 4**	**Income/expense tracking and charts functional**
M5	**User Testing & Bug Fixing**	**Week 5**	**Quality assurance and UAT completion.**
M6	**Final Submission & Deployment**	**Week 6**	**Final project management plan submitted**

Table 2.1

2.2 Scope Statement & WBS

2.2.1 In-Scope

- [ ] Development of a responsive Personal Finance web application
    - [ ] Income and expense tracking with custom categories (Housing, Transportation, Food, Utilities, Entertainment, Savings)
    - [ ] Pie charts and bar graphs for spending visualization
    - [ ] Budget alerts (email and in-app) when nearing category limits
    - [ ] User authentication via email and Google OAuth
    - [ ] Historical reporting with monthly and yearly comparisons
    - [ ] Data privacy and security (HTTPS, data encryption at rest).

2.2.2 Out-of-Scope

- ➔ Google Calendar integration
- ➔ Social media platform integration (Twitter, Instagram)
- ➔ AI-based predictive analytics beyond simple trend lines
- ➔ Real-time stock trading or cryptocurrency tracking
- ➔ Mobile native applications (iOS/Android) - web only
- ➔ Multi-language support (English only for MVP)

2.2.3 Work Breakdown Structure (WBS)

The following WBS is created using Trello for visual task management. It breaks the project into 5 major phases with 25 work packages.

Phase 1: Initiation (Week 1)

Task ID	Task Name	Duration	Dependencies
1.1	Develop Project Charter	3 days	None
.1.2	Identify Stakeholders	2 days	1.1
1.3	Define SMART Objectives	2 days	1.1
1.4	Secure Instructor Approval	1 days	1.2 ,1.3

Phase 2: Planning (Week 2)

Task ID	**Task Name**	**Duration**	**Dependencies**
2.1	**Gather Requirements**	**5 days**	**1.4**
2.2	**Create WBS**	**2 days**	**2.1**
2.3	**Develop Schedule (Gantt Chart)**	**3 days**	**2.2**

2.4	Estimate Costs	2 days	2.2
2.5	Identify Risks	2 days	2.1

Phase 3: Design (Week 2-3)

Task ID	Task Name	Duration	Dependencies
3.1	Create Wireframes	3 days	2.1
3.2	Design Database Schema	2 days	2.1
3.3	Design UI/UX Mockups	3 days	3.1
3.4	Review Design with Stakeholders	2 days	3.2,3.3

Phase 4: Development (Week 3-4)

Task ID	Task Name	Duration	Dependencies
4.1	Setup Development Environment	2 days	3.4
4.2	Develop User Authentication Module	3 days	4.1
4.3	Develop Transaction Entry Module	4 days	4.2

4.4	**Develop Category Management**	**2 days**	**4.3**
4.5	**Develop Data Visualization Charts**	**3 days**	**4.4**
4.6	**Develop Budget Alert System**	**2 days**	**4.5**

Phase 5: Testing & Deployment (Week 5-6)

Task ID	Task Name	Duration	Dependencies
5.1	**Conduct Unit Testing**	**3 days**	**5.6**
5.2	**Conduct Integration Testing**	**2 days**	**5.1**
5.3	**User Acceptance Testing (UAT)**	**4 days**	**5.2**
5.4	**Bug Tracking & Fixing**	**3 days**	**5.3**
5.5	**Deploy to Production**	**2 days**	**5.4**
5.6	**Final Documentation & Handover**	**2 days**	**5.5**

Table 2.2

Work Breakdown of the Above phase

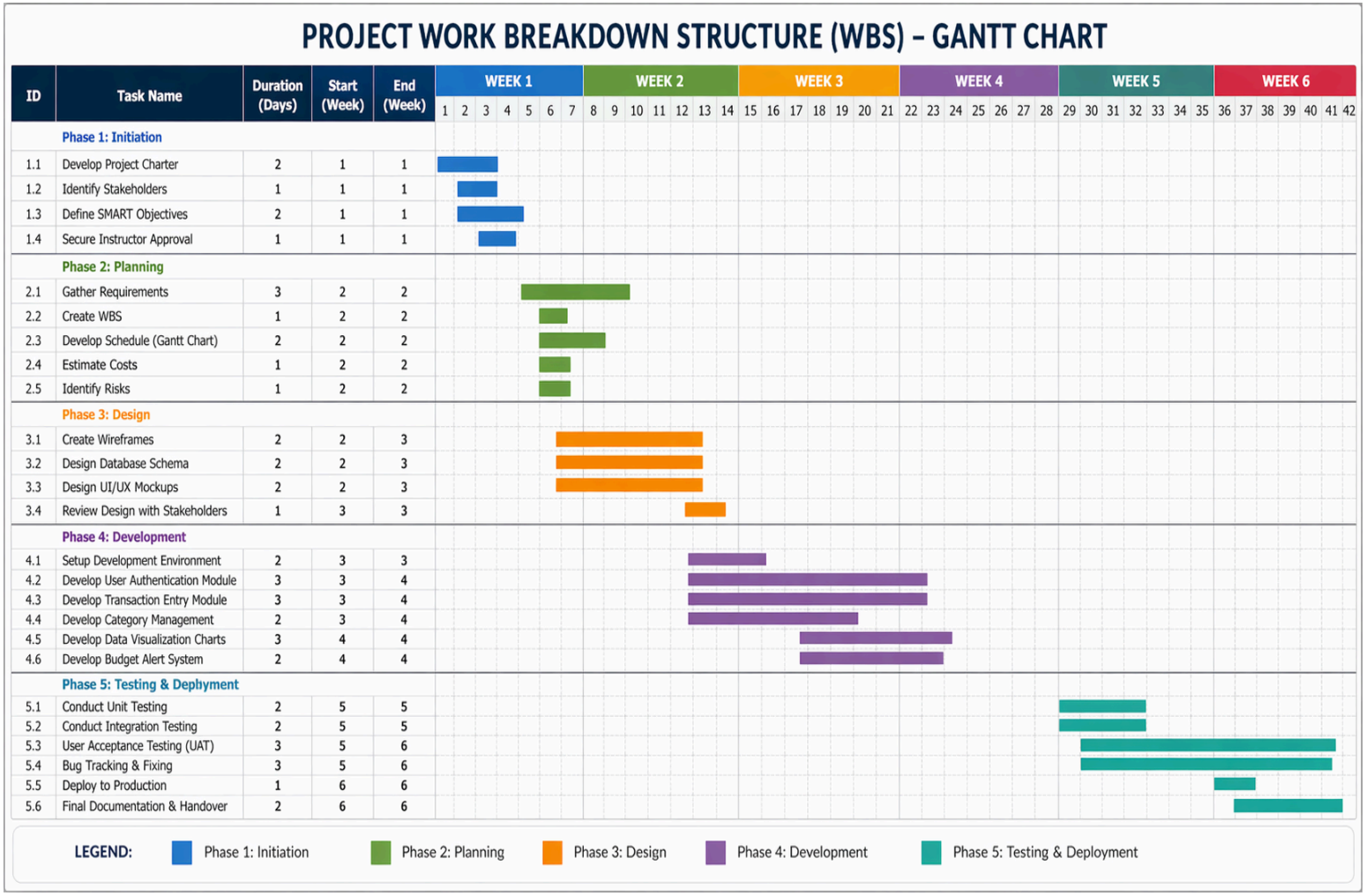


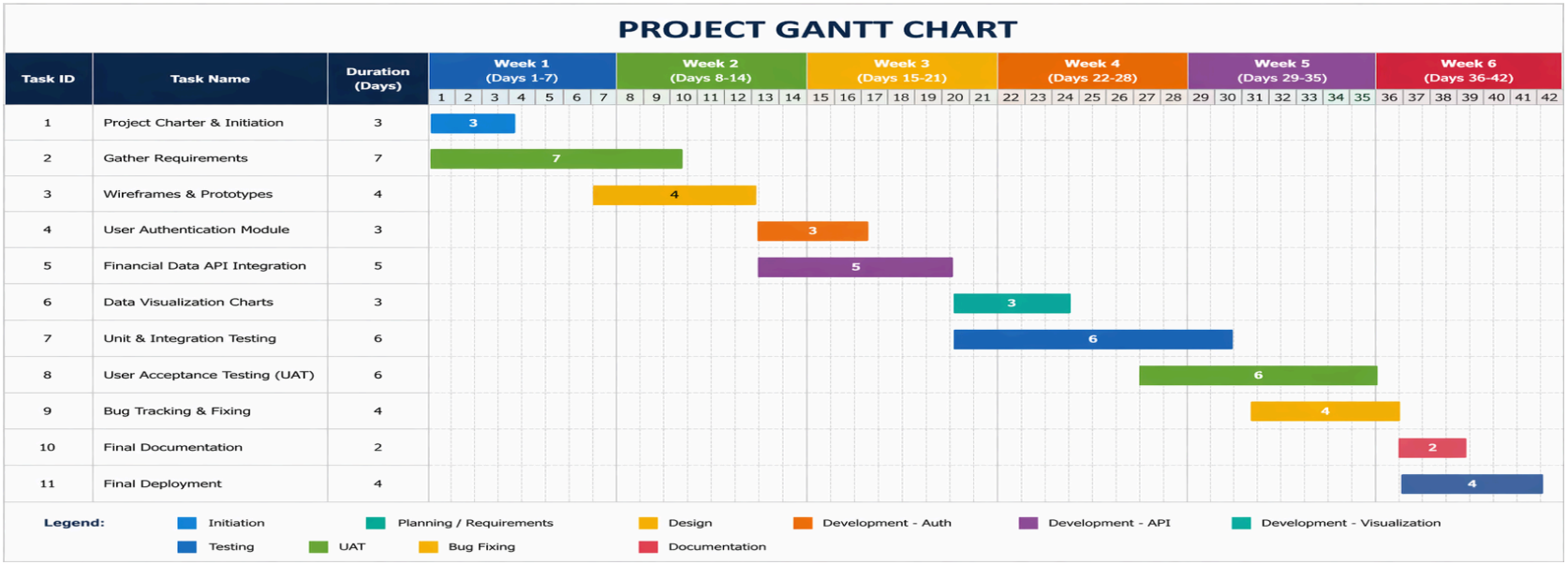
Figure 1

Chapter 3:

Project Schedule

3.1 Gantt Chart Overview

The project schedule is built using Microsoft Project, which allows for precise task dependency mapping and critical path analysis. The Gantt chart below visualizes the sequence and duration of each task, including dependencies such as "Development Environment Setup" must be completed before "Develop User Authentication Module."



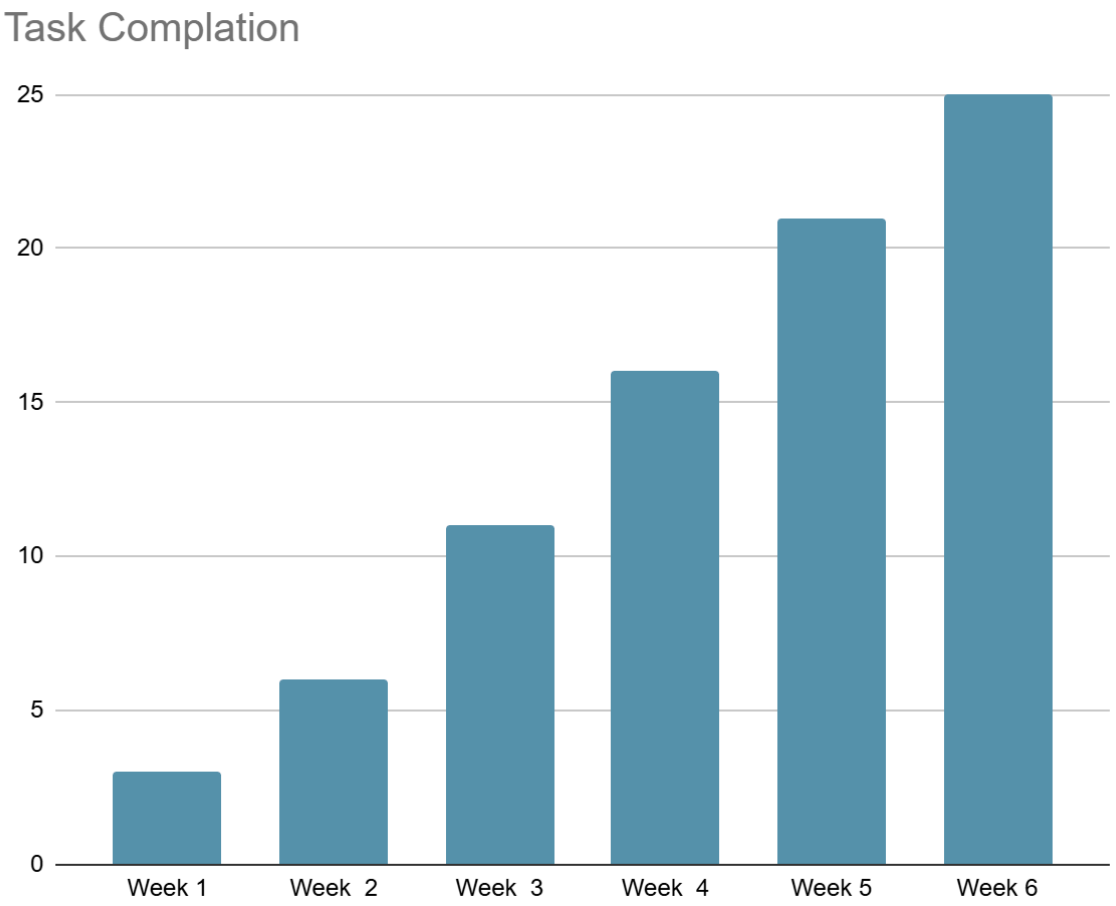

Task Complation
25
20
15
10
5
0
Week 1
Week 2
Week 3
Week 4
Week 5
Week 6

Figure 2

3.2 Task List with Dependencies

Task List and Dependencies

Task ID	Task Name	Duration (Days)	Dependencies
1	Develop Project Charter	3	0
2	Gather Requirements	7	0
3	Create Wireframes & Prototypes	4	1,2
4	Develop User Authentication Module	3	1
5	Integrate Financial Data API	5	4
6	Implement Data Visualization Charts	3	3,5

7	**Conduct Unit & Integration Testing**	**6**	**2**
8	**User Acceptance Testing (UAT)**	**6**	**6,7**
9	**Bug Tracking & Fixing**	**4**	**5**
10	**Documentation & Handover**	**2**	**7**
11	**Final project**	**4**	**8, 9,10**

Table 3.1

3.3 Critical Path Analysis

Critical Path Analysis (CPA) is a project management technique used to identify the longest sequence of dependent tasks that must be completed on time for the entire project to finish by its deadline. The tasks on this path are called "critical tasks" because they have zero "float" or "slack" time.

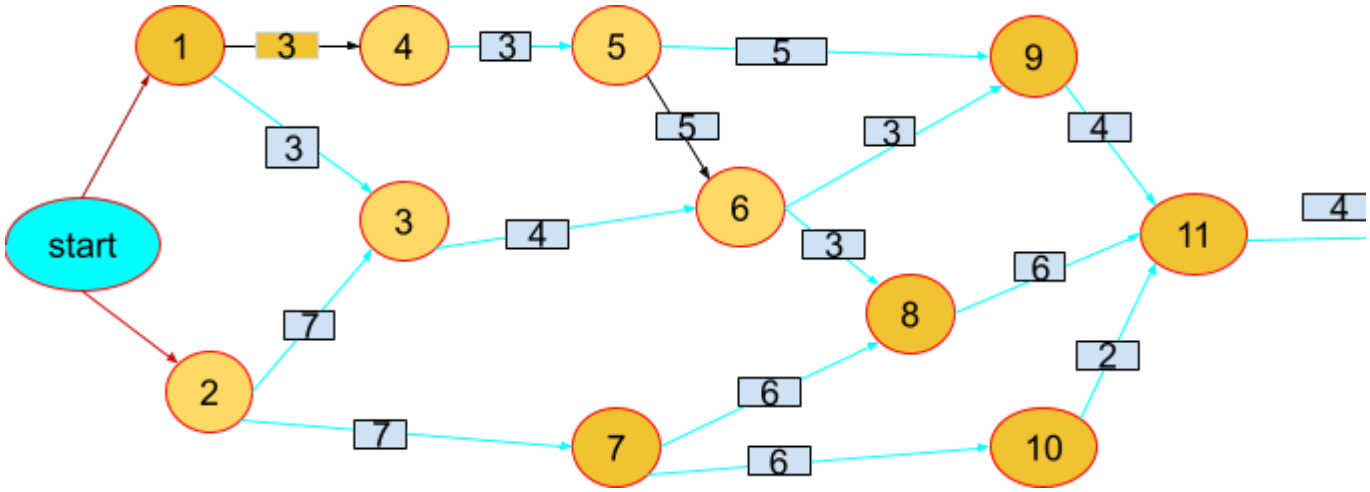


Figure 3

Note : Assume All duration are Days

Path 1: 1-4-5-9-11 Length = 3+3+5+4+4 = 19 days.

Path 2: 1-4-5-6-9-11 Length = 3+3+3+5+4+4 = 22 days.

Path 3: 1-3-6-9-11 Length = 3+4+3+4+4 = 18 days.

Path 4: 1-3-6-8-11 Length = 3+4+3+6+4 = 20 days.

Path 5: 2-3-6-9-11 Length = 7+4+3+4+4 = 22 days.

Path 6: 2-3-6-8-11 Length = 7+4+3+6+4 = 24 days.

Path 7: 2-7-8-11 Length = 7+6+6+4 = 23 days.

Path 8: 2-7-10-11 Length = 7+6+2+4 = 19 days.

The longest path is 24 days.

The longest paths are:

- Path 2: 2→ 3→ 6→ 8→11 (24 days)

Chapter 4:

Cost Estimation

4.1 Detailed Cost Breakdown

The cost estimation is prepared using Microsoft Excel to ensure accurate calculations. The total project cost is 1,456,455 ETB, which includes labor, hardware, software, and a contingency reserve. All monetary values are in ETB.

Cost Estimation Table

Category	Item	Rate (ETB)	Quantity/ Duration	Total (ETB)
Labor	**Project Manager**	**6,500 /hour**	**40 hours**	**260,000**
Labor	**Senior Developer**	**7,800 /hour**	**80 hours**	**624,000**

Labor	**Junior Developer**	**3,900 /hour**	**40 hours**	**156,000**
Hardware	**Testing Laptops (2)**	**78,000**	**2 units**	**156,000**
Hardware	**Smartphone for Testing**	**39,000**	**1 unit**	**39,000**
Software	**Cloud Hosting (AWS)**	**19,500 /month**	**3 months**	**58,500**
Software	**Domain Name**	**1,950**	**1 unit**	**1,950**
Software	**Microsoft Project License**	**3,900 /month**	**3 months**	**11,700**
Software	**Lucid Chart License**	**1,300 /month**	**3 months**	**3,900**
Software	**API Integration Fees**	**13,000**	**1 unit**	**13,000**
Contingency	**10% of Subtotal**	**N/A**	**N/A**	**132,405**

Total Project Budget **1,456,455 ETB**

Table 4.1

4.2 Cost Allocation Chart

A Cost Allocation Chart is a financial management tool used to identify, aggregate, and assign project costs to specific activities, resources, or departments. In the context of the Smart Personal Finance Dashboard project, this chart breaks down the total budget of **1,456,455 ETB** into distinct categories such as Personnel (Labor), Hardware, Software licenses, and Contingency funds. Its primary purpose is to provide transparency, ensure financial accountability, and track expenditure against the planned budget to prevent cost overruns.

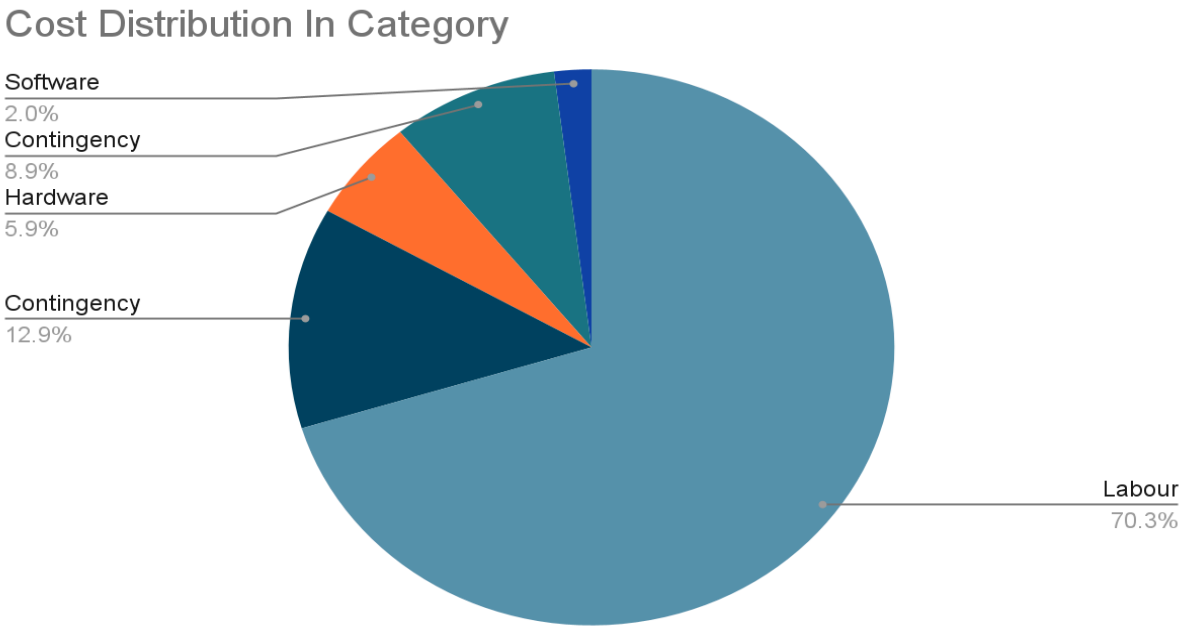


Figure 4

4.3 Cost control

Cost Control for the Smart Personal Finance Dashboard project is to use Earned Value Management (EVM) combined with weekly tracking of actual costs against the approved baseline of 1,456,455 ETB. First, establish a clear cost baseline and break it down by category (labor, hardware, software, contingency). Then, every week, calculate Cost Variance (CV = EV – AC) and Cost Performance Index (CPI = EV / AC) to see if you are over or under budget. Use simple tools like Google Sheets or Excel to record all actual expenses from timesheets, invoices, and receipts. Set a control threshold of ±10%, meaning if any category exceeds its budget by more than 10%, you immediately take corrective action such as reducing overtime, switching to free software alternatives, or delaying non-critical purchases. Always get approval before using any of the 132,405 ETB contingency reserve, and only use it for identified risks like API downtime or hardware failure. Finally, review your cost performance weekly with the project team, update your forecast for the remaining work, and document all variances to improve future cost estimation. This approach is professional, easy to implement, and widely accepted in project management.

Chapter 5:

Quality Management

5.1 Quality Metrics

To ensure the Smart Personal Dashboard meets user expectations, three key quality metrics are defined. These metrics are monitored using automated tools and processes throughout the development lifecycle.

Metric	Target	Measurement Tool	Process
Page Load Speed	**< 3 seconds**	**Google Lighthouse**	**Automated performance tests in CI/CD pipeline**
Defect Rate	**< 5 critical bugs per release**	**Jira Bug Tracker**	**Weekly bug triage and resolution**
User Satisfaction Score	**> 4.5 / 5**	**Post-UAT Survey (Google Forms)**	**Collect feedback after UAT and implement improvements**

Table 5.1

5.2 Quality Assurance Processes

The project will use a combination of manual and automated testing. Code reviews will be conducted using Google Docs for documentation and peer reviews for code. Automated testing frameworks (e.g., Jest for unit tests) will be integrated into the development workflow. Lucid Chart will be used to create process flow diagrams for the QA pipeline, ensuring all team members understand the testing protocol.

Chapter 6:

Communication Plan

6.1 Stakeholder Communication Table

Effective communication is critical for project success. The following table outlines the communication plan, created collaboratively using Google Sheets to ensure transparency and accessibility.

Communication Plan

Information	Stakeholder	Frequency	Method
Project Status Report	Project Sponsor, Instructor	Weekly	Email (Gmail) + Google Docs
Daily Stand-up Updates	Development Team	Daily	Telegram Group
Risk Register Updates	Project Manager, Team	Bi-weekly	Google Sheets (shared)

Table 6.1

6.2 Collaboration Tools

The team will rely on Telegram for instant messaging and quick updates, as it enables file sharing and reminders. Google Docs will be used for collaborative document creation, such as the Project Charter and final report. Microsoft Project will be used for the Gantt chart, which will be shared as a PDF snapshot during weekly status reports. All communication will adhere to a professional tone to maintain clarity and accountability.

- Telegram: Instant messaging
- Google Docs: Documentation
- Trello: Task management
- Google Sheets: Budget & Risk tracking
- MS Project: Gantt chart

Chapter 7:

Risk Register

7.1 Identified Risks

Using a risk register created in Google Sheets, the project team has identified five potential risks. Each risk is analyzed for probability and impact, and a mitigation strategy is defined. The risk register is reviewed bi-weekly to ensure proactive management.

Risk ID	Risk Description	Probability (1-5)	Impact (1-5)	Risk Score	Mitigation Strategy
R1	**API downtime from Fitbit or Google**	**3**	**4**	**12**	**Implement fallback mock data; monitor API status via Telegram alerts**
R2	**Key developer leaves the project mid-way**	**2**	**5**	**10**	**Cross-train junior developer; maintain documentation in Google Docs**

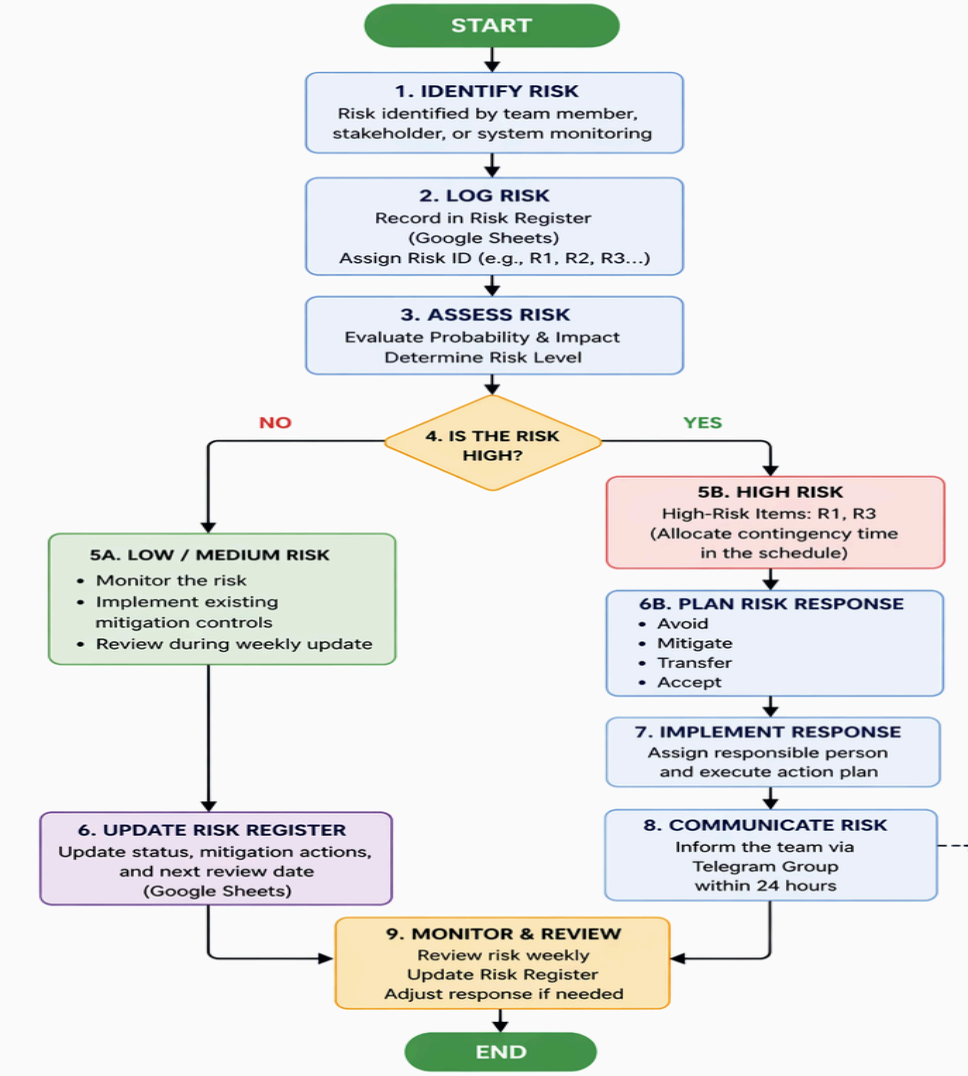
R3	**Scope creep due to feature requests**	**4**	**3**	**12**	**Strict change control process; use Trello to manage backlog**
R4	**Security breach exposing user data**	**2**	**5**	**10**	**Implement HTTPS, data encryption, and regular security audits**
R5	**Incompatibility with target browsers**	**3**	**3**	**9**	**Use cross-browser testing tools; test on Chrome, Firefox, and Safari**

Table 7.1

7.2 Risk Response Planning

For high-risk items (R1 and R3), the team will allocate contingency time in the schedule. The risk register is stored in Google Sheets and updated weekly. The project manager will communicate any new risks via the Telegram group within 24 hours of identification. Lucid Chart will be used to create a risk response flowchart to guide the team in decision-making.

RISK RESPONSE FLOWCHART
Smart Personal Finance Dashboard Project



PROJECT

Smart Personal Finance Dashboard

A web-based application to help users track income, expenses, budgets, and financial goals with interactive dashboards.

RISK LEVEL KEY

- High-Risk (Score 15-25)
- Medium Risk (Score 8-14)
- Low Risk (Score 1-7)

HIGH-RISK ITEMS

R1: Data security breach (User financial data exposure)

R3: Integration failure with banking/financial APIs

TOOLS & COMMUNICATION

- Risk Register: Google Sheets
- Communication: Telegram
- Flowchart Tool: Lucidchart

NOTES

- Risk register updated weekly
- New risks communicated within 24 hours
- Contingency time allocated for high-risk items (R1 & R3)

RISK RESPONSE STRATEGIES

AVOID — Change the plan to eliminate the risk

MITIGATE — Take action to reduce probability or impact

TRANSFER — Shift the risk to a third party

ACCEPT — Acknowledge the risk and monitor

Chapter 8:

Conclusion and Recommendation

8.1 Conclusion

The Smart Personal Finance Dashboard project successfully outlines a comprehensive plan to address the challenges of personal financial management. By integrating essential project management tools such as the Work Breakdown Structure (WBS), Gantt chart, and Risk Register, the team has established a clear roadmap for developing a functional and user-friendly web application. The project is designed to be completed within a 24-day timeframe, ensuring that all core features—from real-time expense tracking to data visualization—are delivered with high quality. This project not only demonstrates technical software engineering skills but also highlights the importance of structured project planning in achieving successful software delivery**.**

8.2 Recommendations

To further enhance the system, it is recommended to integrate machine learning algorithms for predictive budget analysis and to expand the platform into a mobile application for better accessibility. Continuous security audits should also be conducted to protect user financial data.